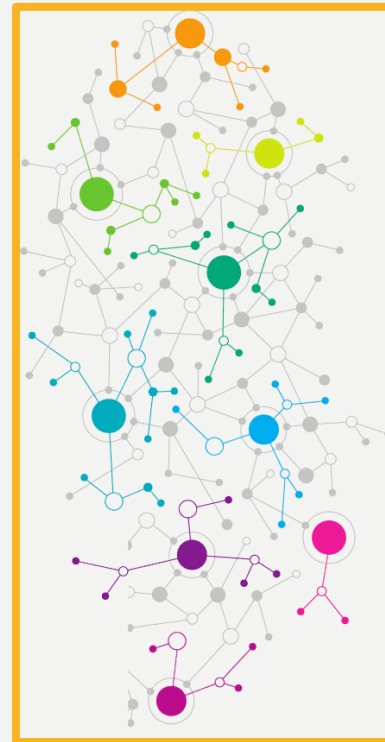




REDES NEURONALES PROFUNDAS PARA RECONOCER IMÁGENES

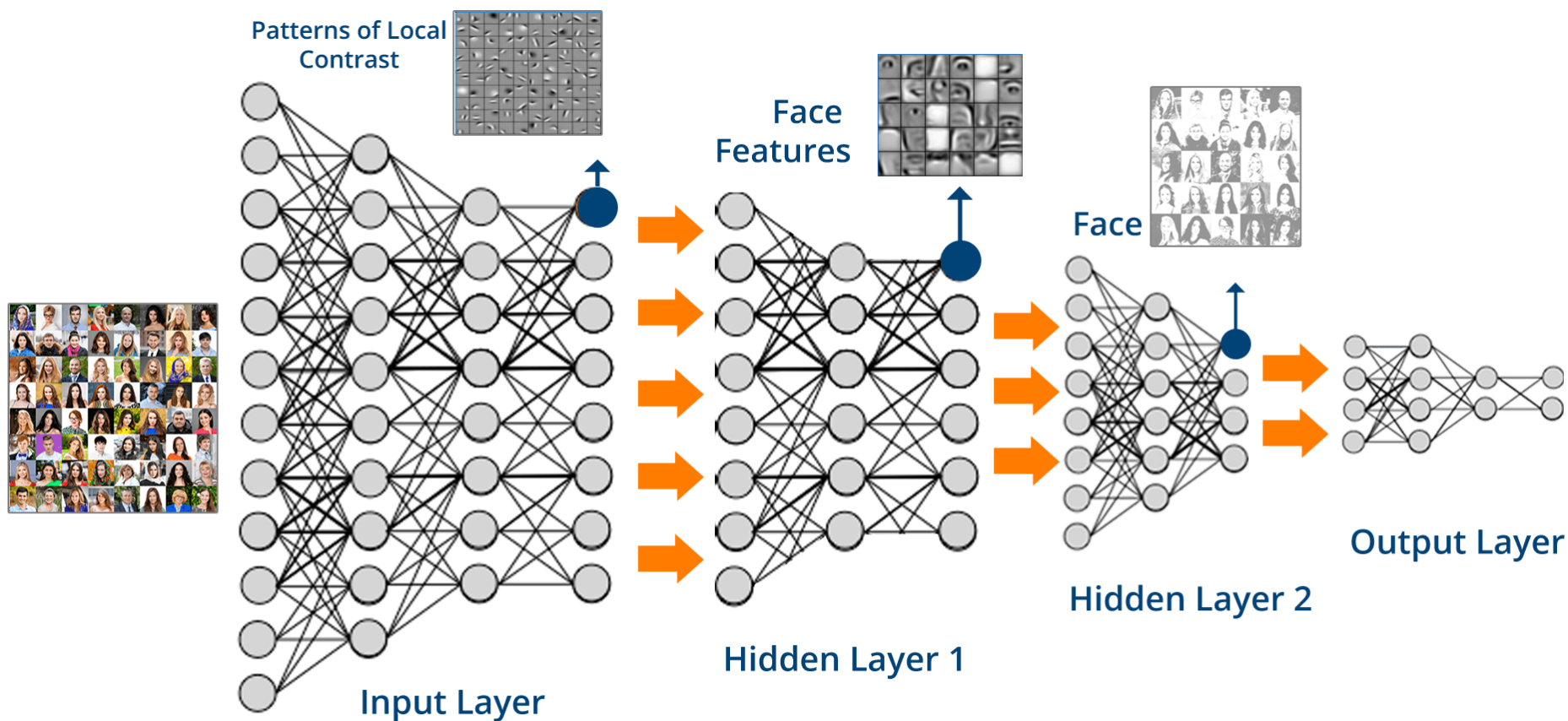
OSCAR JOSÉ PELLICER VALERO

ENTRENAMIENTO



→ ¿Perro / Gato?





REDES CONVOLUCIONALES RECONOCIMIENTO DE CARAS

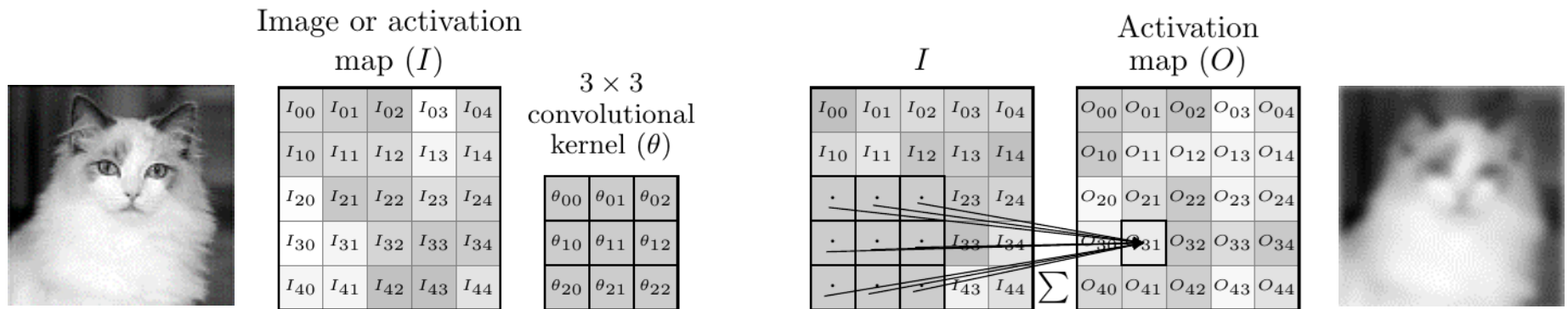
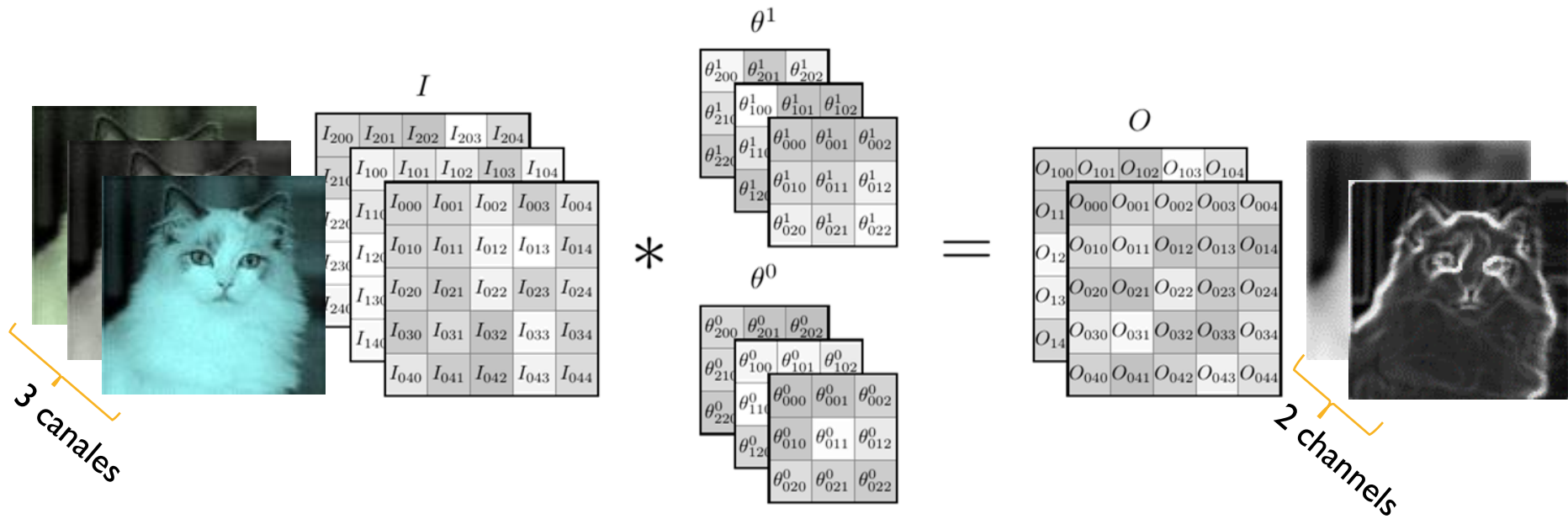


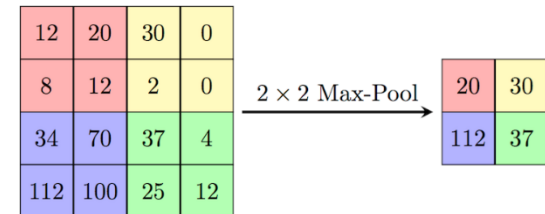
Figure 1.3: Convolution operation on a simple 2D gray-scale image. 2D: Two-dimensional

REDES CONVOLUCIONALES
QUÉ ES UNA CONVOLUCIÓN



$$\text{ReLU}(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ x & \text{otherwise} \end{cases}$$

Max- pooling



REDES CONVOLUCIONALES
QUÉ ES UNA CONVOLUCIÓN

VISUALIZACIÓN DE LA OPERACIÓN DE CONVOLUCIÓN

PARAMETERS

Kernel Size: Stride:

Padding: Dilation:

Output = $\text{floor}((8 + 2 \times 1 - 1 \times (3 - 1)) / 1 + 1) = 8 \times 8$

KERNEL PRESETS

Identity Edge-H Edge-V
Sharpen **Blur** Emboss

INPUT DIGIT

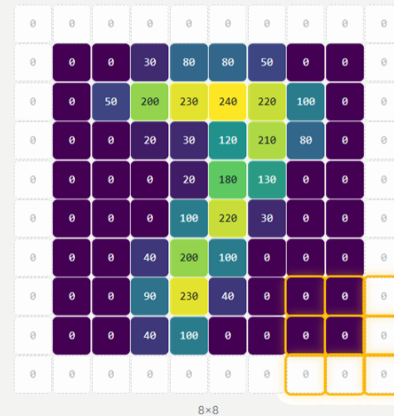
0 1 2 3 4 5 6
7 8 9

ANIMATION

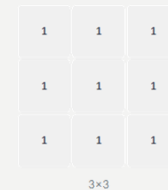
▶ Play ⏸ Pause ⏮ Di Step ⏪ Reset

Speed:
Step 64 of 64

INPUT (8×8)



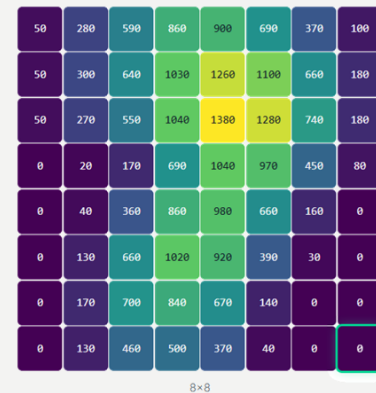
KERNEL (3×3)



×

=

OUTPUT (8×8)



Low High (Input / Output) Low High (Kernel)

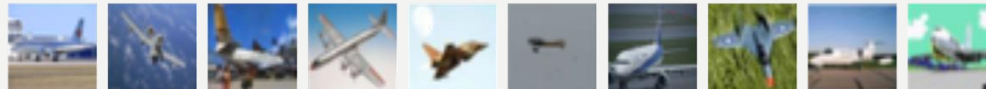
<https://sesen.ai/convolution-visualizer>

APRENDIZAJE PROFUNDO: RECONOCIMIENTO DE ESCRITURA

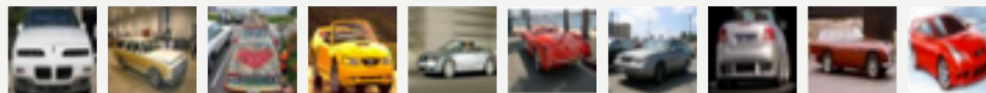


APRENDIZAJE PROFUNDO: RECONOCIMIENTO DE IMÁGENES

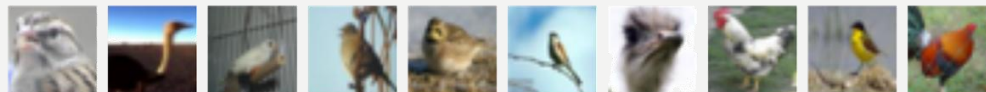
airplane



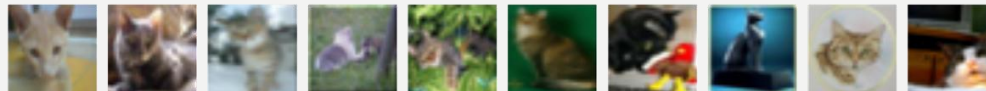
automobile



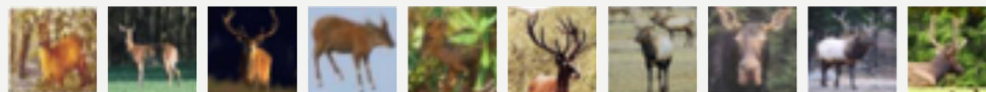
bird



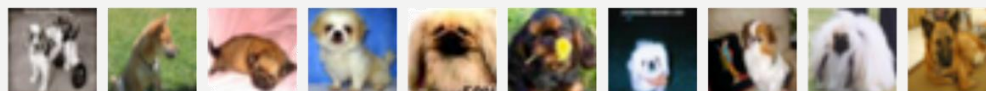
cat



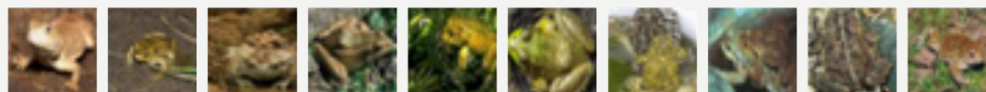
deer



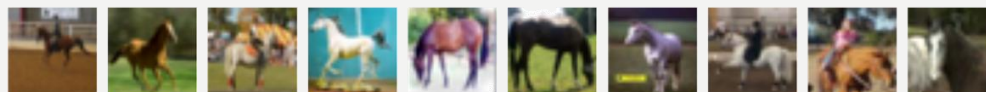
dog



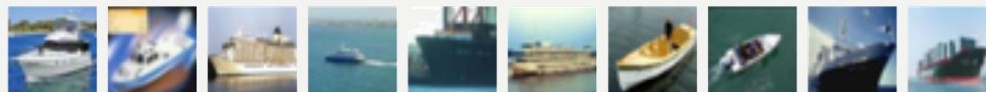
frog



horse



ship



truck

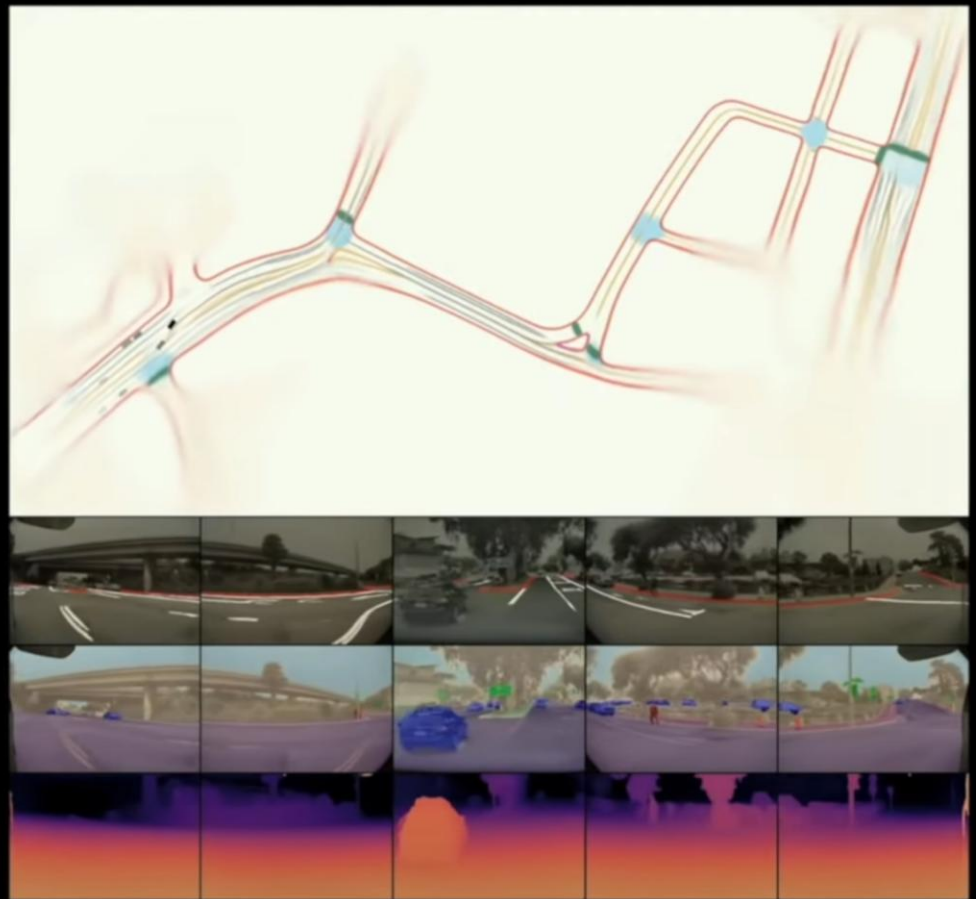
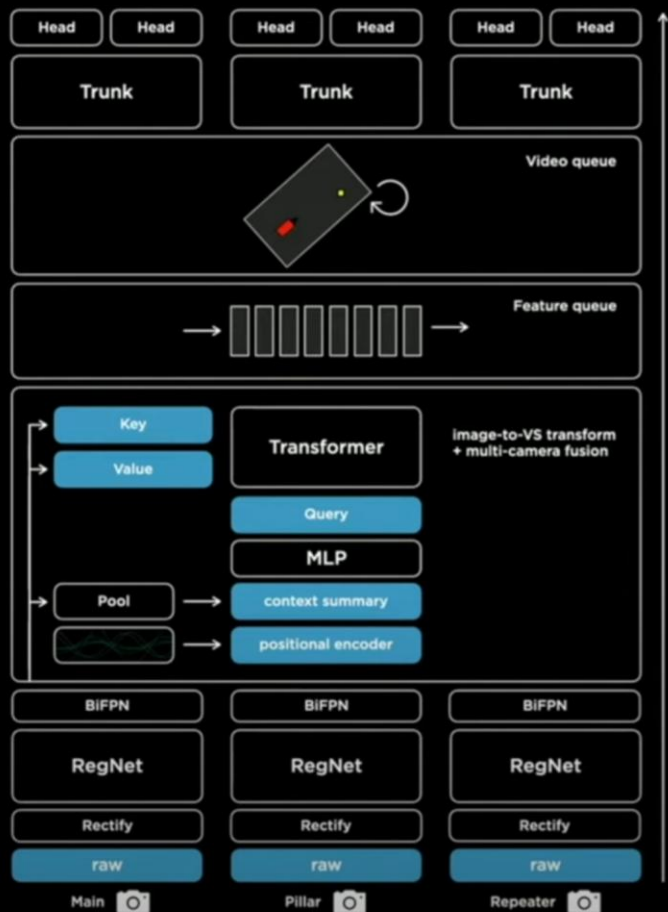


APRENDIZAJE PROFUNDO: DETECCIÓN Y SEGMENTACIÓN DE IMÁGENES



https://www.youtube.com/watch?v=__kxX09i4fds
<https://www.youtube.com/watch?v=0yCYro9HIkM>

APLICACIONES REALES: TESLA



<https://youtu.be/keVVEE9FwS9o?t=96>